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Analysis of the German Commuter Network

Period 2013 - 2021

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Abstract The number of commuters in Germany has continued to rise in recent years. Methods from the field of Complex Networks Analysis (CNA) have also been increasingly used to analyze this phenomenon. The work in progress presented here, shows existing approaches in this area. Based on this, an approach for the analysis of commuter obligations in Germany over the period from 2013 to 2021 is being implemented.

Keywords Commuting · Complex Network Analysis · Spatial Data

1 Introduction

The number of commuters in Germany has continued to increase in recent years. According to the Federal Office for Building and Regional Planning - BBSR, the number of commuters rose by 6.6 percent in five years, from 18.4 million employees in 2016 to 19.6 million employees in 2021. This corresponds

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to 59.5% of all those who are subject to social security contributions. The group of commuters with a one-way commute of more than 50 kilometres experienced a sharp increase of 7.4%. It rose from 3.3 million in 2016 to 3.6 million in 2021. Among the major cities, Munich continues to lead the list with the most commuters [1].

2 Literature

There is already some work on the use of CNA for the analysis of commuter networks with a wide variety of objectives. He et al. developed a method to identify statistically significant communities in a weighted network with a high proportion of self-looping weights. The regional delineation approach presented a method for inferring geographic regions systematically from the underlying data using community detection methods in network science [2].
De Montis et al. also studied the structure of the network representing the inter-municipal commuting traffic of the Sardinia region, Italy. They calculated different topological properties of the network, degree, shortest path length, degree distribution, variance, clustering coefficient, degree similarity of the neighbours of a node and additional weighted measures like probability distribution of the weights, strength of a node, probability distribution of the strength, disparity, the weighted clustering coefficient and a local weighted average of the nearest neighbour degree [3]. A follow up approach to the last mentioned work was also presented by De Montis et al.. The analysis utilized measures like the degree, probability distribution of the degree, average and maximum path length between any pair of nodes, clustering coefficient, efficiency and different weighted measure like probability distribution of the weight and strength [4].
Patuelli et al. presented a first approach for the analysis of the spatial commuting network in Germany over time. Several properties have been taken into account during the analysis process like clustering coefficient, weighted clustering coefficient, diameter, average degree, standard deviation degree, max degree, min degree, centralization, betweenness mean, betweenness std. dev. and max betweenness. The study results suggest that the commuting network appeared to have become more dense and clustered over the period considered (1995 – 2004) [5]. The afore mentioned approach was continued by Nijkamp et al. for the comparison of the German commuter network for the years 1995 and 2005 utilizing a weighted network also based on districts. The observed measures were selected according to three perspective, centralization, clustering and variety/dispersion and contained, e.g. indegree centralization, visibility, relative indegree centrality, aggregated network indegree centralization, (k-order) clustering coefficients and entropy, whereas the k-core concept was introduced for the analysis of subgraphs. The results showed that, in addition to a local multi-nodal commuting network (between nearby cities), a regional network was also present to some extent — which, however, did not overshadow well-defined local relations [6].

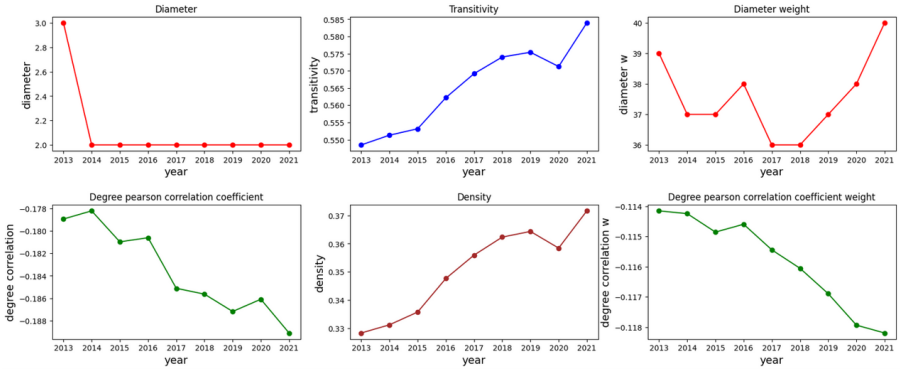

Fig. 1 Global Parameters

3 First Results for the network analysis on German commuter data

The goal of the ongoing work presented here is to analyse the commuter flows in Germany over the period of 2013 to 2021. Using all years of the period makes it possible that short term effects due to the corona pandemic could be taken into account. The created network consists of the German districts (nodes) and the in commuter flows (Einpendler) (edges) and incorporates weights based on the overall number of commuters. In commuter flows from outside German districts were excluded.

First results of the analysis conform with the afore mentioned findings of the BBSR regarding the increase in commuting activity. For example an analyse of the minimum degree showed an increase from 30 in 2013 to over 40 in 2021. The median degree increased from 115 in 2013 to more than 130 in 2021. The degree distribution showed that most districts have a low number of degrees while there are a few districts having a high number. Figure 1 shows an overview how different measures evolved over the period. The analysis of the community structure of the network also revealed interesting insights. Figure 2 shows the change of the community structure over the period computed using the Louvain Community Detection Algorithm. The communities fit the geographical distribution quite well except for some larger cities like e.g. Berlin and Hamburg which were assigned to the south eastern cluster. We can also see that the cluster in central Germany disappears in 2014, 2019 and 2021. The analysis of the modularity showed low values about 0.2 with a slight negative trend.

4 Availability of data and materials

The data representing the commuter flows was provided by the Federal Employment Agency, BA (Bundesagentur für Arbeit) (Dataset "Arbeitsmarkt in Zahlen, Sozialversicherungspflichtig Beschäftigte - Pendler nach Kreisen"). It

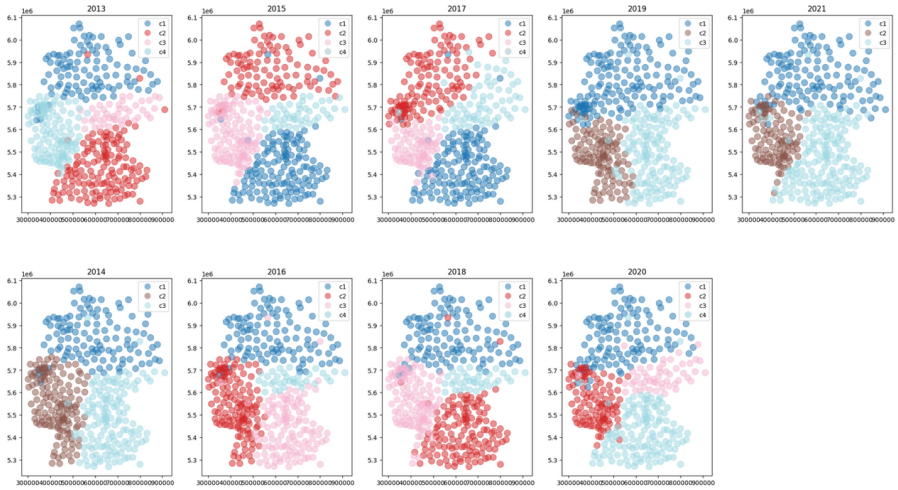

Fig. 2 Community structure

should be stated that there are different annotations in the dataset regarding the data quality concerning the years 2013/14, 2016/2017 and 2019 due to technical issues or local government reorganization. Coordinates were added to the dataset using the dataset Geographische Namen 1:250 000 (GN250) (© GeoBasis-DE / BKG 2021) provided by the Federal Agency for Cartography and Geodesy (BKG).

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